



GLOVE AND MESH APRON WASHING MACHINE

Manufacturer :	ROSER Construcciones Metálicas S.A
Machine/Equipment:	Glove and Apron Washing Machine
Code/ Ref.:	13319 - RA00000171
Serial Nr.:	****
Voltage:	380V 3PH 50/60Hz
Manual Nr.:	13319-2013-eng



DECLARACIÓN DE CONFORMIDAD DECLARATION OF CONFORMITY

ROSER Construcciones Metálicas, S.A.
Ctra. Girona a Sant Feliu, nº 369 – Km. 11,500 - 17244 Cassà de la Selva (Girona) - España
Teléfono : (34) 972 460434 - Fax : (34) 972 463700

La empresa **ROSER Construcciones Metálicas, S.A.**, declara bajo su responsabilidad que la máquina,
Company **ROSER Construcciones Metálicas, S.A.**, declares under its own responsibility that,

- Tipo/ type : MAQUINA DE LAVAR GUANTES Y DELANTALES DE MALLA.
- Nº de Serie/ Serial N°: ***
- código/ code: 13319 - RA000000171
- Año de construcción/ year 2014

instalada de acuerdo con las reglas profesionales, normas de instalación y las instrucciones del fabricante,
debidamente mantenida y utilizada en las aplicaciones para las que se destina, cumple con los requisitos de las
directivas correspondientes que le son aplicables según,

provided that it is installed, maintained and used in the application for which it is made, with respect of the
professional practices, relevant installation and manufacturer's instructions,

- Directiva de máquinas **2006/42/CE**.
- Machine's directive

- Directiva de Baja Tensión **2006/95/CE**
- Low tension's directive

- Directiva de Compatibilidad Electromagnética **2004/108/CE**
- Electromagnetic compatibility's directive

es conforme a la normativa.
it is in rules conformity.

Cassà de la Selva, a 15 Febrero de 2014

Lluçia Casellas Puig
Gerente - General
Manager





1 – TABLE OF CONTENTS

1 – TABLE OF CONTENTS	9
4 – ACERCA DE ESTA EDICION	13
4.1 – REVIEWS AND ISSUES	13
4.2 – TECHNICAL ASPECTS	13
4.3 - CONTENTS	13
5 – SAFETY NORMS	14
5.1 – GENERAL WARNINGS ON SAFETY	14
6 – CONDITIONS OF USE AND NORMS	16
6.1 - ANTECEDENTS	16
6.2 – DANGER WARNINGS	16
6.3 – AIR NOISE AND VIBRATIONS	16
7 – GENERAL DIMENSIONS	17
8 – TECHNICAL SPECIFICATIONS	18
9 - RECEPTION	19
9.1 – BEFORE DOWNLOAD	19
9.2 - DOWNLOAD	19
9.3 – TRANSPORT INDOORS	19
10 – INSTALLATION	20
10.1 - POSITION	20
10.2 - CONNECTIONS	21
11 – INSTRUCTIONS FOR USE	23
11.1 – RUNNING-IN	23
11.2 – EQUIPMENT DESCRIPTION	23
11.3 – PROCESS	25
12 – CLEANING AND MAINTENANCE	26
12.1 - MAINTENANCE	26
12.2 - TOOLS	26
12.3 – MAINTENANCE INSTRUCTIONS	26
12.4 - CLEANING	27
12.5 – CLEANING INSTRUCTIONS	27
12.5.1 – CLEANING PRODUCTS	28
12.5.2 – CLEANING OF NOZZLES	28
12.6 – SPARE PART PACKAGE	29
13 - INCIDENTS	30
13.1 - ANOMALIES	30
13.2 – TROUBLESHOOTING	30
14 – EXPLODED VIEW	31
14.1 – Exploded view. Lower drum rail	32
14.2 – Shutting system. Front Door	32
14.3 – Exploded view. Front door shut sensor	33
14.4 – Exploded view. Drive	33
15 – LIST OF PARTS	34



16 – POWER+CONTROL-SCHEMATIC R1863.....	36
LIST OF COMPONENTS - R-1863.F	41
ANNEX : Manual Zelic Logic 2. Configuration parameters	42



2-WARRANTY

1 - ROSER Construcciones Metàlicas S.A. guarantees its installations, machines and equipment for a period of **12 months** from the installation date, but not later than **18 months** a from the delivery date of the machines and/or equipment. This date will be indicated in our delivery note or international transport documentation.

2 - ROSER Construcciones Metàlicas S.A guarantees that all the equipment and machines have been tested and verified working correctly before its dispatch.

3 – A necessary condition for the repair under **Warranty** is that the equipment and machines shall be repaired by **ROSER Construcciones Metàlicas S.A**, or by the authorized dealer and using original spare parts with the manufacturer's instructions and that the **damaged parts** be sent in a period of **two** days after the damage. It is indispensable to return the damaged parts so that the **Warranty** can be applied.

4 - The **Warranty** includes the repair or exchange of the parts of the product that cannot be used due to manufacturing or mateiral defects.

5 – The manufacturer's **Warranty** will finish if the buyer resells the machine after its use.

6 - The **Warranty** will not be applicable in case of damages due to an unsuitable installation, unadequate service, overload, usual wearing, non-suitable accesory installation or misuse.

7 - The **Warranty** does not cover the damages to people or other assets caused by the machine.

8 - The **Warranty** does not cover damages caused by external causes, or damages caused at the same time or indirectly by damaged components.

9 - The **Warranty** does not cover the working time loss, waiting or installation time, or transport damage that generate realted with the repair.

10 - The **Warranty** does not cover any mechanical component if manufacturing or material defects are found.

11 - The **Warranty** does not cover any electrical component if manufacturing or material defects are found.

12 - ROSER Construcciones Metàlicas S.A reserves the right to introduce modifications or improvements in its equipment or machines without prior notice; being not obliged to modify the machines and/or equipment so far produced.

1 - PREFACE

We thank you for the trust on **ROSER Construcciones Metàlicas S.A** when purchasing one of our products.

The objective of this documentation is to advise you about the steps that shall be followed from the reception of the machine, its download, indoors transport, installation, commissioning and running-in, as well as the correct maintenance for a long useful life.

It is recommended to follow the advices and warnings in this text, since its working life will be depending on its installation, use and conservation.

We inform you that our commercial and technical services are always at your request for any event or demand that might be requested. We also thank beforehand your collaboration to help us improve the present document.

¡¡ VERY IMPORTANT !!

For the technical requests as well as to order spare parts, it is necessary to indicate:

The Project number indicated in the general drawing of the installation “OT-000000” and the position number of the machine.

Additionally to the description of the machine or equipment, other identification data, such as: (those found in the characteristic plate)

Manufacturing date:

Model/Code:

Serial number:



Part of the information referred is found in the identification plate that each machine or equipment has with it, in which we indicate the mentioned parameters, as well as other characteristics.

We shall also indicate the code or ref. of the parts that are requested; these codes can be found in the tables that are described in the different part list in the corresponding section of the manual or the machine.

For any additional Information or demand, we ask you to contact the SAT (post-sales service) or our Technical Service, addressing yourself to:

ROSER Construcciones Metàlicas S.A ESPAÑA

Ctra Girona a Sant Feliu, nº 369-Km 11.5 -17244 -Cassà de la Selva (Girona) -España

Tel.: (34) 972 460 434 - Fax: (34) 972 463 700

Correo electrónico (e-mail) : roser@roser.es - Web <http://www.roser.es>

You can also obtain Information and solutions through your closest dealer.



4 – ACERCA DE ESTA EDICION

4.1 – REVIEWS AND ISSUES

Reference nr. of this manual: **13319-2013**

Data: Issue: March 2013

The new issues are complete reviews of the manual. The modification sheets, that can be published between two issues, contain additional Information. The issue dates can only change when publishing a new issue.

Those minor corrections that do not affect to the function of the product can be implemented in a further reprint without changing the Printing date.

A certain number of updates and amendments of the product do not require to modify the manual and, in the same way, the corrections in the manual can be carried out without that implying to carry our modifications of the product. We might not expect, therefore, to have a direct correspondence between the updates of the products and the reviews of the manual. I.

4.2 – TECHNICAL ASPECTS

The information included in this document belongs to its owner and is protected by the copyright laws.

All the rights reserved. Any part of this document shall not be photocopied, reproduced or translated to any other Language, nor yielded to third parties, without the accepted consent in written by **ROSER Construcciones Metálicas, S. A.**, according to the corresponding legal liabilities.

The instructions and recommendations, as well as the information described in this manual, shall be exclusively used for the machine(s), equipment or installation(s) the aforementioned manual is dealing with.

ROSER Construcciones Metálicas, S. A. reserves the right to carry out the modifications considered needed, being possible that it modifies, supersede or include, unilaterally and without prior notice, new contents and/or services, as well as the way these appear presented and positioned..

4.3 - CONTENTS

ROSER Construcciones Metálicas, S. A. carries out the utmost effort to avoid any error in the contents that could appear in this manual. We do not guarantee, nor we can be consider liable of the consequences that could derive from the errors in the contents that could appear in this manual, given by third parties. In the same way, it does not assume itself liable for any of the discrepancies that could exist between the printed documents and the electronic version published in our website. In case of discrepancy between the printed version and the one in the website, the printed version will be aplicable. .

ROSER Construcciones Metálicas, S. A. can not be hold accountable for the damages, indirect or derived, that can be produced with respect to the supply, efficiency or bad use of this document and the instructions, recommendations and information therein described.



5 – SAFETY NORMS

ii ATTENTION !!

To all the buyers, industrial safety directors, inspectors, users, operators and people in charge of maintenance, repair and cleaning. Carefully read the instructions and precautions for safety before installing, using, running-in, repairing or cleaning any equipment.

5.1 – GENERAL WARNINGS ON SAFETY

The operators shall be instructed according to the user manual. They will be instructed as well about norms and regulations on work safety and environment, being demanded to comply with those. It shall periodically verified that the operators work according to the instructions of use, with knowledge on the possible threats and applying the established safety directives.

In the case that the operator observes any anomaly, the machine shall immediately be disconnected and warning shall be given to the maintenance staff.

The machine described in this documentation has been designed for maximum safety for the operator and to be compliant with the aplicable laws; however, it has to be pointed out that certain uses and inadequate manipualtions can provoke accidents, causin damages to the equipment and injuries to people in certain specific areas.

To highlight some points or instructions, we use the following símbolos in this manual:



- This symbol is used to **warn** about certain safety precautions that represent a hazard when not followed.



- This symbol is used to warn about certain safety precautions that represent an electrical shock hazard when not followed.



- Symbol that is used to indicate the **duty** by the user to comply with the specified instructions in order to guarantee the safety in the operation and/or protection of the equipment



- This symbol is used to **inform** prompting on specifications and/or instructions that shall remember and/or apply.

People's safety shall be everybody's responsibility and precaution. A copy of this manual shall be available to anyone using, running, Servicing, repairing or cleaning the machine that is described in it. The text will need to be read in detail and trying to understand it well. The safety instructions will be explained in detail to those that are not capable to read and understand the text.

It is possible to obtain more copies of this manual contactin our **Technical Service**.

!! VERY IMPORTANT !!

Most of the machines and equipment used in the meat processing industry imply risks due to high voltaje, parts that move at high speed, Sharp tolos, and other characteristics on devices to cut and elaborate meat. The protections are installed to offer a higher safety level for the operator.

The machines shall not be modified in any case, nor make them run if there are parts pending, parts that are damaged or are not working adequately; we shall do the same when dealing with safety devices. Furthermore, they will not be used for any other purpose than those for which they were designed and assigned.

For your own safety, we list here a number of precautions that shall always be taken into account:

* In the case that the machine has protection elements, either machanical, as caps, covers, windows as well as protections, or electrical, for diferent areas on it, we shall act accrfodngly taking into account that:

- We shall never disassemble nor bypass these protections to introduce hands or tools in those areas, without the machine being stopped and disconnected from the electrical supply.

- The machine shall never run without the protection elements being positioned correctly and verified with the guarantee to ensure safety when running.

- Wes hall never eliminate the electrical safety nor the mechanical safety under no circumstances.

* Do not manipulate, neither try to repair electrical elements, inside or outside the electrical panel, without having completely disconnected the electrical supply.

* Do not position, neither leave tools, parts, etc. Inside the electrical panal, or in the owrking places, nor close to them, to avoid any dangerous situation.

* Do not use parts that are not assigned to this machine, nor exchange them for other diferent than these.

* Carry out the necessary reviews and periodic maintenance of the machine, to have the right and correct work of the machine, expandint its useful life and maintaining the adequate protection and safety level.



- Keep free the area straight adjacent to the equipment. Keep out people that will not intervene on its operations. The non-compliance with these instructions and safety precautions might originate injuries.

!! ATTENTION !!

ROSER Construcciones Metálicas, S.A. cannot be considered accountable for any type of accident that results provoked by any mechanical, electrical change or modifications of performance or positioning in the machine without that being communicated to **ROSER Construcciones Metálicas, S.A.** and thereafter approved by our technical service.



6 – CONDITIONS OF USE AND NORMS

6.1 - ANTECEDENTS

The washing machine has been designed to perform in an automatic and continuous way the washing and rinsing of crates and/or trays. It can work as a stand-alone system or as a part of a cleaning or hygiene line/process.

The equipment integrates the washing by means of a recirculation of hot water and the rinse by means of use of tap water (open loop or recovered), with a crate and/or tray transport system by means of roller chain and with adjustable speed.

The washing machine has a buffer tank for water with an integrated system for the automatic control of the level and a filtering system for water recovery. It also has a centrifugal pump for water recirculation.

For possible uses different from those of the foreseen use for the machine described, it is needed to ask our [Technical Department](#) on each case.

6.2 – DANGER WARNINGS

The equipment shall always be at all moments correctly levelled and fixed firmly to the ground, being needed to review those anchoring points checking that the position and anchoring are correct.

Under no circumstances, neither protections nor safety components in the equipment shall be manipulated or bypassed, since this equipment **shall not** work without the right conditions of use.

It is recommendable to carry out a preventive and corrective maintenance that ensures the perfect performance of the equipment; the operations that derive from the realization of these and that require a manipulation in the equipment, such as repair or cleaning tasks, shall be carried out under the safety conditions suitable for the staff assigned to it, being necessary thus to totally disconnect the equipment from main electrical supply.

The equipment shall not work in case of breakdown or until it has been repaired and verified for its use. The repairs shall always be carried out by qualified professionals and rigorously accomplishing with the applicable norms and regulations.

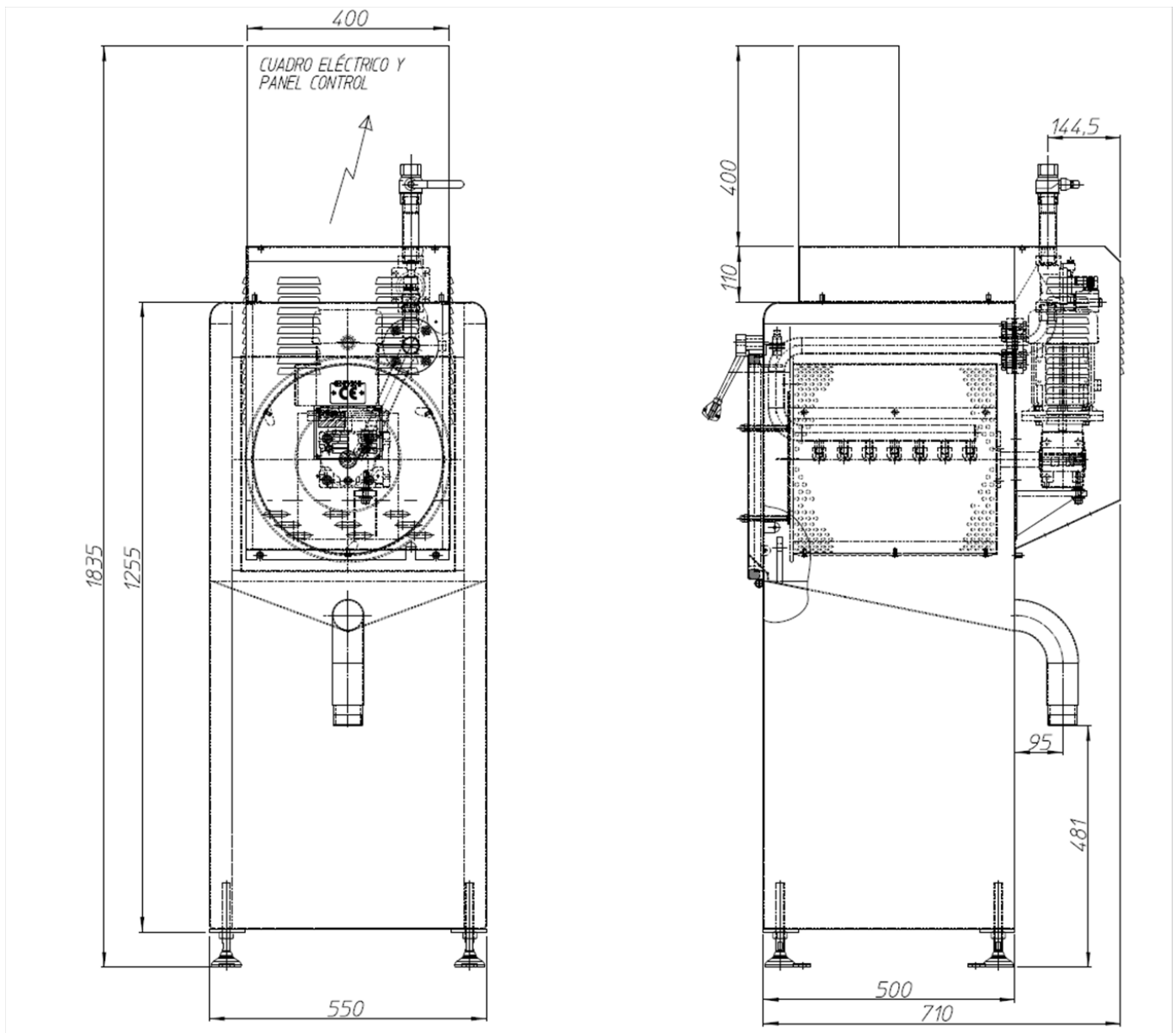
The operator shall avoid any unforeseen maneuver that could provoke an inadequate or unsafe performance of the equipment, avoiding any risky or dangerous situation.

6.3 – AIR NOISE AND VIBRATIONS

The sound intensity emitted by the equipment during its performance is on a very low level, being less than 80 dB (A).

The level of vibration that has been checked during the work of the system, can be established far below the value ($a_{eq} = < 0,5 \text{ m/s}^2$), and it has to be pointed out that this noise is totally dampened by the whole equipment.

7 – GENERAL DIMENSIONS



Supplies

Water inlet: G-1" – Male thread

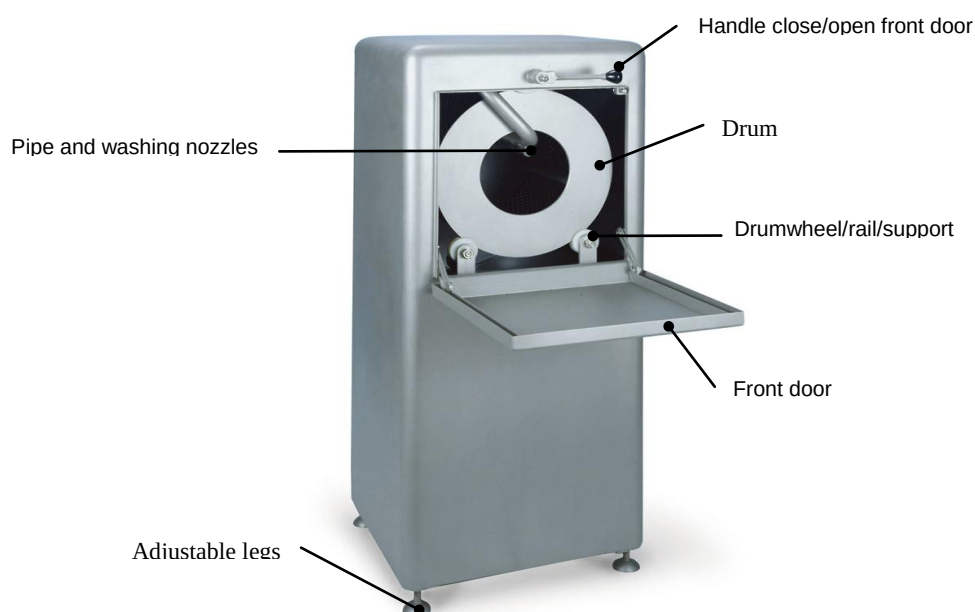
Drain: G- 2" – Male thread

Electrical supply : 380V - 50/60Hz – 3Ph + TT



8 – TECHNICAL SPECIFICATIONS

Description of the machine:	Glove and Metal Apron Washing Machine
Material:	In Stainless Steel AISI 304
Energy requirements:	Water and electrical current supplies
Supplies:	<i>Electrical Supply:</i> Cabinet for connection 380 V – 3Ph - 50/60 Hz + TT <i>Water Supply:</i> Male thread G-1”
Working pressure:	<i>Water:</i> 2 - 5 Kg. /cm ² .
Electrical control:	24 V. AC.
Process:	Washing system by means of rotating drum and water infeed by spray shower. <i>Washing water temperature:</i> Depends on water infeed. <i>Nº nozzles:</i> 7
Control system:	<i>Safety:</i> inductive detector at the door. <i>Washing process:</i> Timer and electrovalve.
Loading system:	Front. Front door with safety switch.
Washing capacity:	20 gloves or 5 metallic mesh aprons every 2 minutes
Drain:	Male thread G-2”
Electrical power:	0,5 HP / 0,37 Kw
External dimensions :	550 x 775 x 1835 mm.
Equipment weight:	Approx. 110 Kg.
Acoustic pressure level:	Lower than 50 dB (A) absolute
Vibration level:	Value aeq = < 0,5 m./s ²





9 - RECEPTION

9.1 – BEFORE DOWNLOAD

Before the reception of the equipment in the user's premises, it is very important to carry out the necessary checks to verify that the equipment is in perfect state and has not suffered any hit or damage during the transport.

Any anomaly found shall be communicated by written to the transport company and informing on this respect **ROSER Construcciones Metálicas, S. A.** or his representant.

To download, transport, installation, running-in, use or maintenance of the equipment, it is recommended to follow the instructions of this documentation.

9.2 - DOWNLOAD

It is a relatively simple operation, since the dimensions of the equipment do not imply a difficulty, being it possible to easily transport and manipulate it.

Proceed to download supporting the equipment in an adequate way, taking into account the weight and the center of gravity for suspending or holding it, both if the operation is done by means of a crane in the factory, crane in the truck or a forklift; in each case the suitable fixing points will be available foreseen to perform the transport and positioning of the equipment in the right way, without resulting in a dangerous operation itself, even though it is recommended not to realize sudden movements and always taking the right precautions to avoid accidents.

9.3 – TRANSPORT INDOORS

For the transport indoors and the positioning of the machine, it is necessary to take into account and review the following points:

- 1 - The volume dimensions of the equipment and the available space for the transport.
- 2 - The circuit that will follow from the position of download to the position where it will be installed, checking the spaces, and considering the total dimensions (width, height and length), for its transport.
- 3 - The correct position of the connections, both of the machine to the assigned supplies and to the electrical mains.
- 4 - The drains:
 - a. If they are free, check the slopes on the floor and the channels and/or sewers assigned.
 - b. If they have to be channeled, being aware of their position and dimensions will ensure their correct connection.

10 – INSTALLATION



- Before installing the machine, read the instructions and the safety precautions.
- Disregard the instructions and safety precautions can create serious injuries.
- Not connecting the machine to the electrical supply until it has been completely installed.

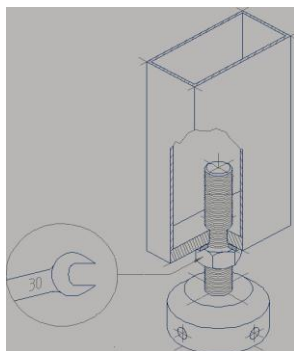
10.1 - POSITION

Place correctly the equipment having available the suitable space and taking into account the needs for the use and operation of the equipment, the position of the connections to the different energy supplies (electricity, steam, water and drains), the cleaning and maintenance operations, ergonomic criteria considering the dimensions of the machine and the required space, as well as the compliance with the norms that may be applicable in each case.

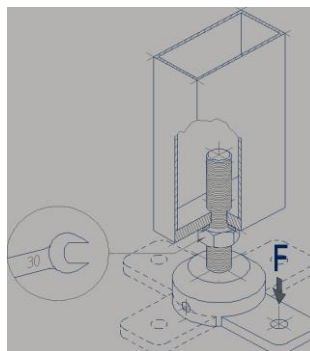
Place the equipment on a flat surface and modify the position of the legs until it is levelled. Loosen the nuts of the adjustable legs on the base, turn them right or left, if we wish to lift it or lower it. Once levelled, fix the nuts again. In case of having feet with fixation tab, it is recommended the use of a fixation in accordance with its type of material. In most of cases it is recommended to fix by use of fixation bolts, either with metallic or chemical anchoring, even though sometimes a fixation with screws is possible.

Base feet of Ø50, screw and hexagonal nut M16 – Key nr. **28** – Torque according to table.

Base feet of Ø60, screw and hexagonal nut M20 – Key nr. **30** – Torque according to table.



Detail **D** – Foot without fixation



Detail **D** – Foot with fixation **(*)****(**)**

(*) The tab will be oriented to the required position

()** It will be fixed to the ground thru **F** with anchoring M10

10.2 - CONNECTIONS



- The connections shall always be carried out by qualified staff and following all the applicable norms and regulations.

10.2.1 – ELECTRICAL CONNECTION

Check that the voltage indicated in the plate of characteristics of the equipment matches the supply to which it will be connected.

Positioned below the electrical panel, we will find the thru connector to connect to electrical supply with a wire of the adequate length and section.

IMPORTANT: To avoid a voltage drop, the wire with which the electrical supply Connection is carried out will have the adequate section in accordance to its length, and without any doubt, will be determined by the position of the equipment with respect to the assigned electrical supply point.

¡ IMPORTANT !

- If the voltage input is changed, the connections of motors and transformer(s) shall be changed and adapt accordingly the circuit-breakers and thermal relays.
- **All the electrical interventions shall be carried out by qualified staff.**
- Modifications of any type are not allowed, without being previously approved by the technical department of **ROSER Construcciones Metálicas, S.A.**
- All the circuits and connections shall be fully compliant with all the electrotechnical low voltage and regional regulations.

The supply connected to the equipment shall be fully compliant with all the applicable low voltage and regional regulations, being necessary the adequate protection.

¡ IMPORTANT !

- To correctly connect the machine to the electrical supply, check that the input voltage is equal to that on the sign positioned to the Connection cable. The line connected to the equipment shall have available:
 - Circuit-breaker - Differential protection
 - Adequate cable section - Grounding

Table Electrical wiring colours

POWER			CONTROL		
Black	Black		220/240 V.AC	Phase = Brown	
Phase (S)	Brown		48 V.AC	Phase = Grey	
Phase (T)	Grey		24 V.AC	Phase = Orange	
Phase (N)	Blue		24 V.DC	(+) Positive = Red	
Ground (TT)	Yellow			(-) Negative = Black	
	Green				



10.2.1.1 – CONTROL PANEL VOLTAGE

La maniobra se realiza **24 V. AC** a través de un transformadora de maniobra. The control is carried out @ 24V A.C. through a control transformer.

10.2.2 – WATER SUPPLY

The connections for the water infeed and drain are located in the rear side of the machine. To reach them, lift vertically the rear cover and extract it.

It is needed a water supply with a pressure in the range from 1 to 3.5 bars (Kg/cm²).

To connect the water supply there is a male thread G-1”.

The Connection has to be secure and watertight, both in the case that the machine is installed permanently as if it is carried out on a temporary position.

The water supply connected to the equipment has to have the suitable pressure and flow for the proper performance of the equipment, as well as a general key that will allow opening or shutting the supply. It shall also have a filtering system at the source to avoid the flow of particles that could clog the nozzle holes.

The efficiency of the equipment can diminish if another equipment that consumes water is connected to the supply pipe and could reduce its performance.

10.2.3 - DRAIN

Mail thread G-2” free output, with the possibility to conduct both if required.



The supplies to which the equipment might be connected, have to have a main key to shut or open the supply, as well as the suitable pressure and flow for the correct performance of the equipment. (See technical specifications, Pages 23 and 29).

11 – INSTRUCTIONS FOR USE



- Before running the equipment, read the instructions and safety precautions.
- The non-compliance with the instructions and safety precautions can originate injuries.

11.1 – RUNNING-IN

11.1.1 – EQUIPMENT INSPECTION

Once positioned and connected to the electrical and water supplies, the equipment shall be inspected and we shall verify that the installation has been carried out correctly.

Check that the covers and protections are correctly positioned, as well as the safety devices, if the equipment has them.

Remove any loose object that might be on the equipment or around it, and maintain the adjacent zones around it, clean and set.

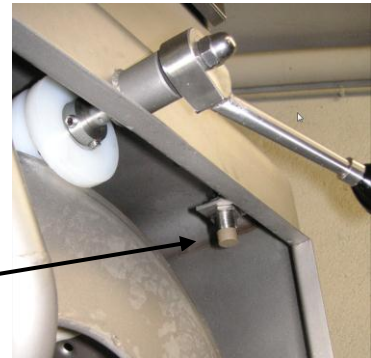
Check the the turning direction of the drum is correct in accordance with the direction of the arrow on it. In the case that the turning direction is not correct, disconnect the equipment from supply and exchange the two phases at the input in the cabinet and check again the turning direction of the drum.

11.2 – EQUIPMENT DESCRIPTION

11.2.1 – Safety system

The machine has a safety switch on the door, not running the machine if the front door is not completely closed.

INDUCTIVE SENSOR



11.2.1 – PLC

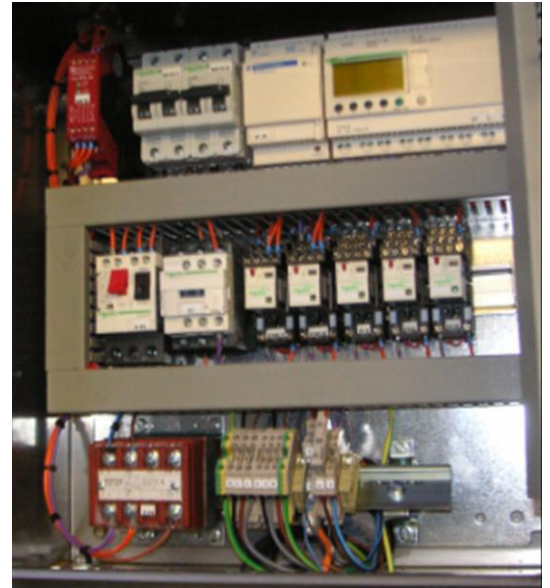
Inside the control cabinet, there is a PLC module (see attached user manual). This is programmed from factory, so that the duration of the washing process be 1 minute approximately.



The times programmed in the factory are :

- T1** – Pre-washing time 8 seconds
- T2** – Emptying time 8 seconds
- T3** – Water infeed time 8 seconds
- T4** – Washing time 8 seconds
- T5** – Rinsing time 8 seconds
- T6** – Draining time of 8 seconds

To modify those settings, the PLC module has to be reprogrammed. We must access the sub-menu of parameters and establish the new parameters. For more Information, please read the manual in the annex.



11.2.2 – Washing

The washing of gloves and mesh aprons is carried out by time by means of a rotating drum and open loop water spray shower.

Drum with capacity for 20 gloves or 5 aprons.

The protrusions inside the drum make the gloves or aprons turn in order to ensure an optimal cleaning operation.



11.2.3 – Drum driving system

Driving system by means of a gear-motor with direct transmission to the drum with an output speed of 34.5 rpm. This is placed in the rear part of the machine with a screwed cover.

11.3 – PROCESS

A – Before starting the process

- Check that the front door is correctly shut
- Regulated washing cycle of 1 minute duration. In case that it should be modified, access the electrical cabinet.

B – Process

- Position inside the drum the material to be washed.
- Activate the machine using the *Pushbutton (PT)* of the switch cabinet. The *ON (PL) light* (green) is on.
- Completely close the front door. The washing cycle starts (the drum turns and water gets in).
- Open the front door when the washing cycle is completed.
- Exchange the already cleaned material by another to be cleaned.
- When shutting the front door, a new washing cycle starts.

WARNING: Two cycles in a row cannot be carried out, since the machine requires that the door is opened and closed to start a cycle.

C – When the working day is finished.

- Completely empty the material from the machine.
- Push the Pushbutton (PT) in switch cabinet in OFF position.



- Do not leave the machine connected to mains when no operator is taking care or when it shall not be working.

12 – CLEANING AND MAINTENANCE



- During the Maintenance, cleaning, or before any intervention on the equipment or machine, select the correct working mode when it is foreseen. Otherwise, always disconnect from mains shutting the supply.

12.1 - MAINTENANCE

The Maintenance works shall only be carried out by qualified staff.

Ensure that all the staff read the complete user manual.

ROSER Construcciones Metálicas, S. A. is not responsible for the accidents and damages that could be caused as a consequence of the given instructions.

12.2 - TOOLS

Use the proper tools for the Maintenance and repair tasks.

12.3 – MAINTENANCE INSTRUCTIONS

Due to its simplicity, this machine does not require a complex Maintenance program for its components.

It is under user responsibility to establish a suitable maintenance program for the use of the equipment.

12.3.1 – SPRAY NOZZLES

It is important to carry out an inspection of the nozzles for the correct operation of the machine, so that we avoid a bad performance of the machine when the nozzles are clogged.

The inspections will be carried out with a higher frequency when the use of the machine is higher.

12.3.2 – GREASING AND LUBRICATION

The gear motor is delivered greased with long duration grease.

Periodically grease the bearing support by means of greasers available. It is recommended to add grease in lower quantity and frequently. To access that bearing, the rear cover has to be taken out. This bearing is located between the gear motor and the machine frame.

12.3.3 – PREVENTIVE MAINTENANCE

In the following table, there is a verification list for the preventive Maintenance, covering such table the main components of the tunnel. This list will be useful to establish a maintenance program.



COMPONENTS	SUGGESTIONS	FREQUENCY		
		Weekly	Monthly	Trimester
Motors Gear-motors	Verify noise		•	
	Verify temperature		•	
	Verify oil level			•
Bearings Supports with bearings	Verify noise		•	
	Lubricate		•	
	Verify connections (screws, supports, etc.)			•
Structure	General verification (screws, supports, etc.)		•	



- *Never remove the protections available in the equipment when this is in operation. Always position again correctly those protections after carrying out the required adjustments.*

12.4 - CLEANING

It is under the responsibility of the user to establish a cleaning program suitable for his needs. This program shall take into account all the applicable laws, the regulations and the standards corresponding to public health and safety for the use and manipulation of chemical products.



- *There could possibly appear eventually stains in some parts of the machine due to friction with other metallic elements, during the installation or as a result of the manipulations when carrying out adjustments or Maintenance. It is recommended to rub these oxidized areas or stains with a scouring sponge (or similar) to eliminate the iron particles attached to the stainless Steel (Never use steel wool or metallic brush).*

It is important to maintain the working area in suitable conditions of cleanliness and order, needed requirements for a better performance both of the equipment and the process to follow, without forgetting that it will be, without a doubt, an improvement in preventing possible risks.

12.5 – CLEANING INSTRUCTIONS

When carrying out the cleaning operations of the equipment using a water jet, always avoid to aim at the electrical components (electrical cabinet, motors, electro valves, etc.).

Never take out the protection covers of the transmission and the motor, since they shall always be protected from water.

12.5.1 – CLEANING PRODUCTS

Clean with water and neutral soap. Never use chlorine detergents, since those attack the stainless Steel, producing colouring and even holes.

On the side of the machine, we have placed a sticker to inform on the possible causes of corrosion, as indicated below:

ATENCIÓN

Este equipo está construido con materiales AISI 304 (18.8)
Cet équipe est construit avec matériel AISI 304 (18.8)
This equipment is constructed with AISI 304 (18.8)

No cubre la corrosión en los siguientes productos:
Ne protège pas la corrosion dans les suivants produits:
Will rust on contact with the following products:

Acido Bromhídrico	Acido Fluorhídrico	Acido Sulfurico
Acido Clorhídrico	Acido Oxálico	Acido Clórico

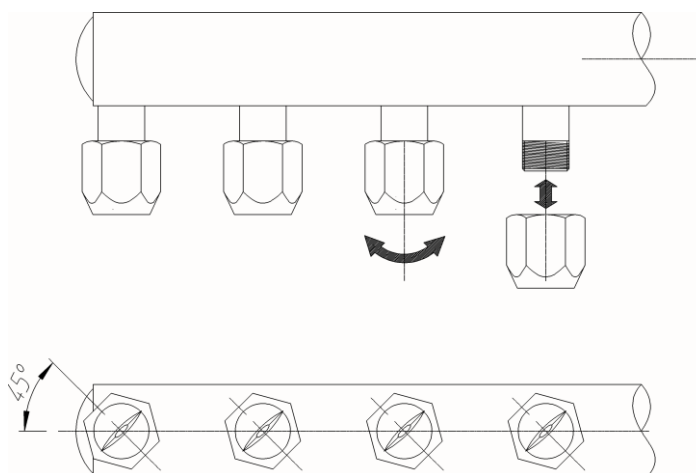
Resto de productos TABLA DE RESISTENCIA QUIMICA DE CORROSION
Le reste de produits dans TABLE DE RESISTANCE CHIMIQUE DE CORROSION
Other product on the CHEMICAL CORROSION RESISTANCE TABLE

12.5.2 – CLEANING OF NOZZLES

If the nozzles become clogged, clean them properly using a rod of the proper diameter, introducing it to unclog them.

It is recommended to use a blowing system thereafter to eliminate the debris or particle that could remain.

Unscrew and separate as indicated here below:



Mount carefully the spary nozzles, so that they have the configuration indicated in the drawing.

12.5.3 – DRUM CLEANING

At the end of the daily work, it is recommended an exhaustive cleaning of the inside of the drum to ensure optimal conditions in the cleaning process.

12.6 – SPARE PART PACKAGE

In order to have a proper Maintenance and to have a fast intervention in possible repairs, we recommend to have some parts or essential components available. This will help in a fast recovery from a breakdown, thus avoiding long and unnecessary stop sor other negative consequences that could be originated by a lack of prevention of any anomaly or breakdown.

12.6.1 – MECHANICAL COMPONENTS

Part Nr.	Units	Description
10589	4	Spray nozzle for washing, G-3/8"
19149	1	Solenoid valve housing G-1"
20051	1	Lower wheel, drum rail, ø60/22,3 x 30 mm
20052	1	Upper wheel, drum rail, ø75/55/20,2 x 30 mm
20067	1	Inner paddle of the drum.

12.6.2 – ELECTRICAL COMPONENTS

Part Nr.	Units	Description
19144	1	Coil for solenoid valve
19145	1	Connector for solenoid valve coil
29167	1	Inductive proximity sensor IGT200



The parts indicated are parts and components that are a portion of the recommended spare part package for maintenance.

13 - INCIDENTS



- Both the installation and the repairs have to be carried out by qualified technical staff and always being compliant with the applicable regulations.
- For any anomaly that is not envisioned in this incident section, it is needed to get in contact with our Technical Service.

13.1 - ANOMALIES

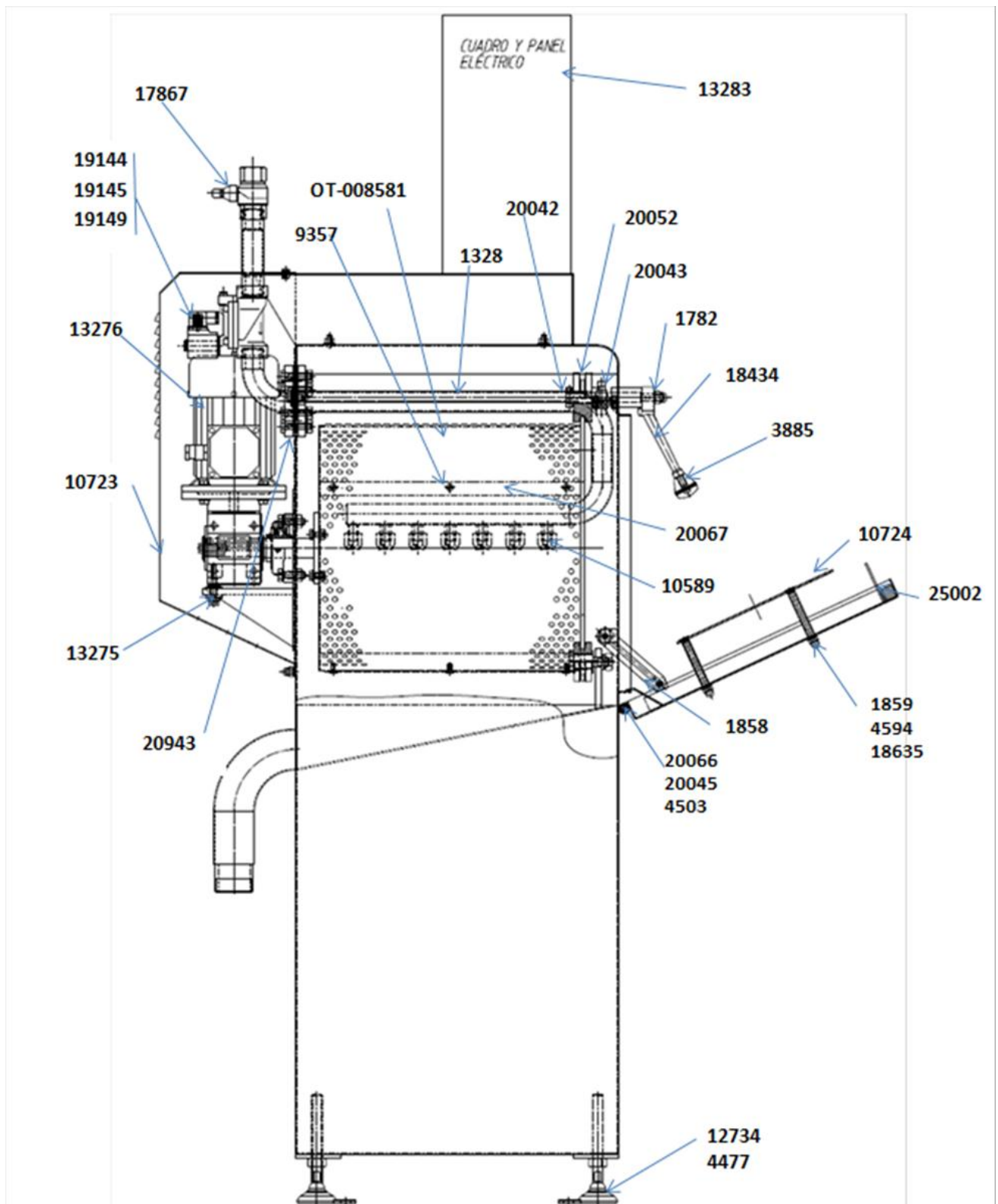
Any equipment or machine suffers with time anomalies or breakdown, either because of the wearing of the components, misadjustment or aging, always related to the complexity of the machine and the use of it, as well as the cleaning, service and maintenance received.

13.2 – TROUBLESHOOTING

The possible anomalies in operation are indicated below, with possible facts, causes and actions to be taken in each case by the maintenance department.

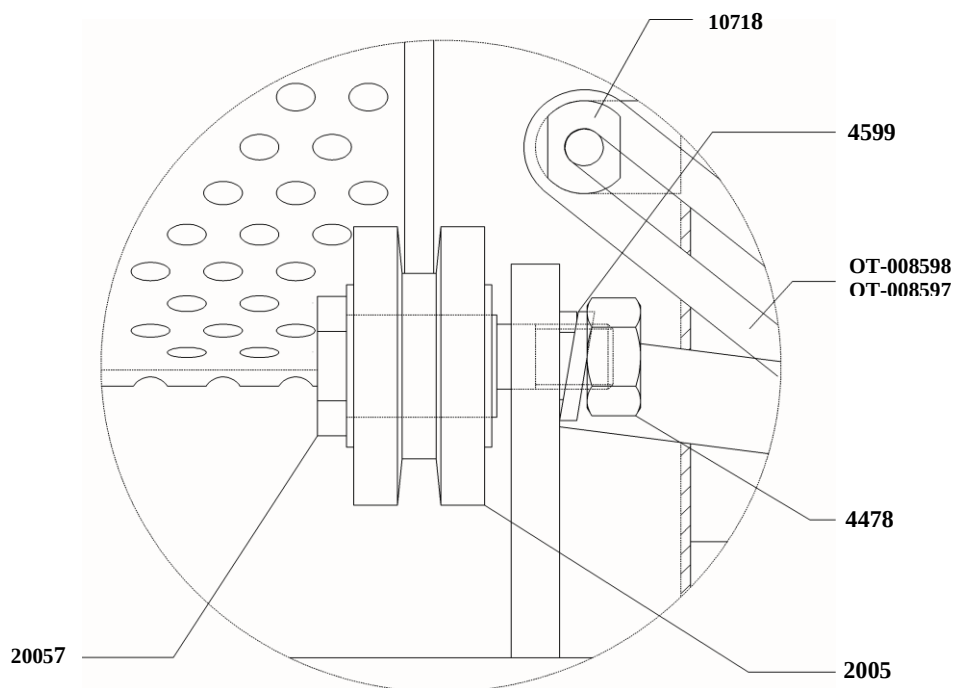
FACT	DUE TO ...	ACTION
➤ The machine does not run	▪ Failure in electrical supply.	- Check White Pushbutton On is activated. - Verify Supply / Reestablish supply.
	▪ Front door not completely closed.	- Close completely front door.
➤ Scarce or No water in the washing stage	▪ Failure in water supply.	- Verify supply/ Reestablish water supply
	▪ Clogged nozzles.	- Check (verify, clean, reposition).
	▪ Failure in electro valve.	- Check (verify, clean, reposition).
➤ Uncorrect/No turn of the drum	▪ Breakdown in bearings.	- Verify connections and supports (tighten/exchange) - Verify bearings (Grease/exchange)
	▪ Breakdown in gear motor.	- Verify its correct performance or repair.
	▪ Wheels of the drum rail worn out.	- Verify and exchange if needed.

14 – EXPLODED VIEW

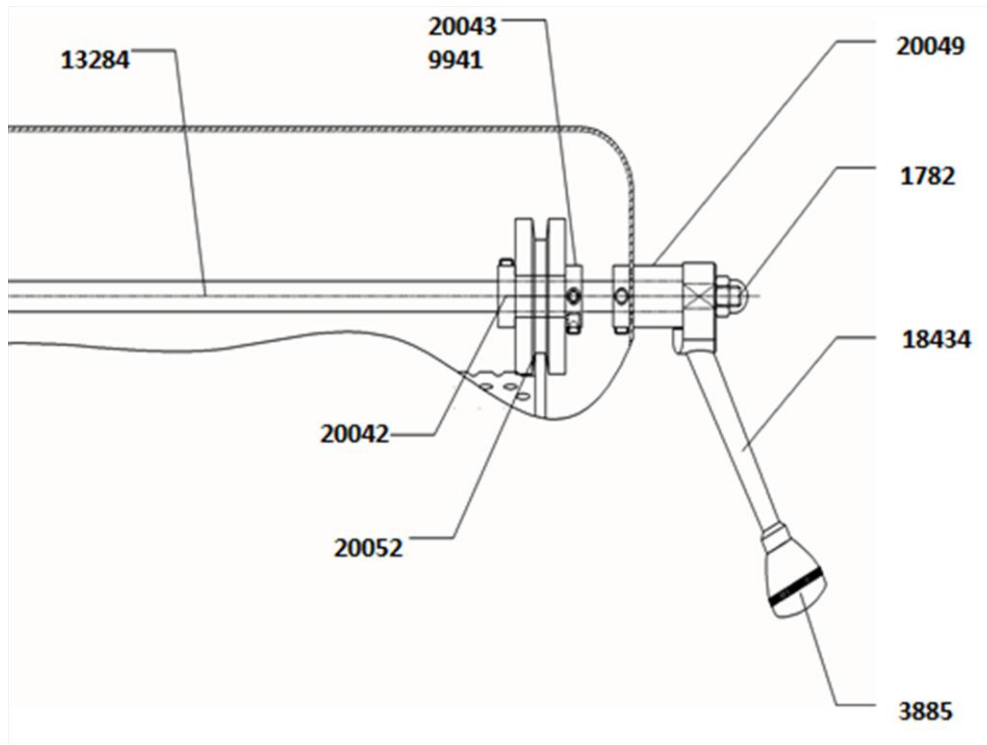




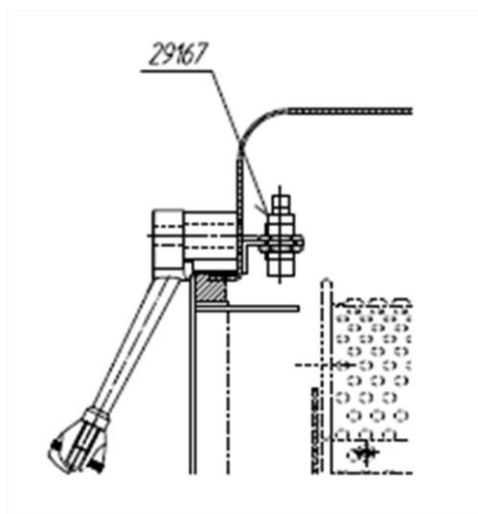
14.1 – Exploded view. Lower drum rail



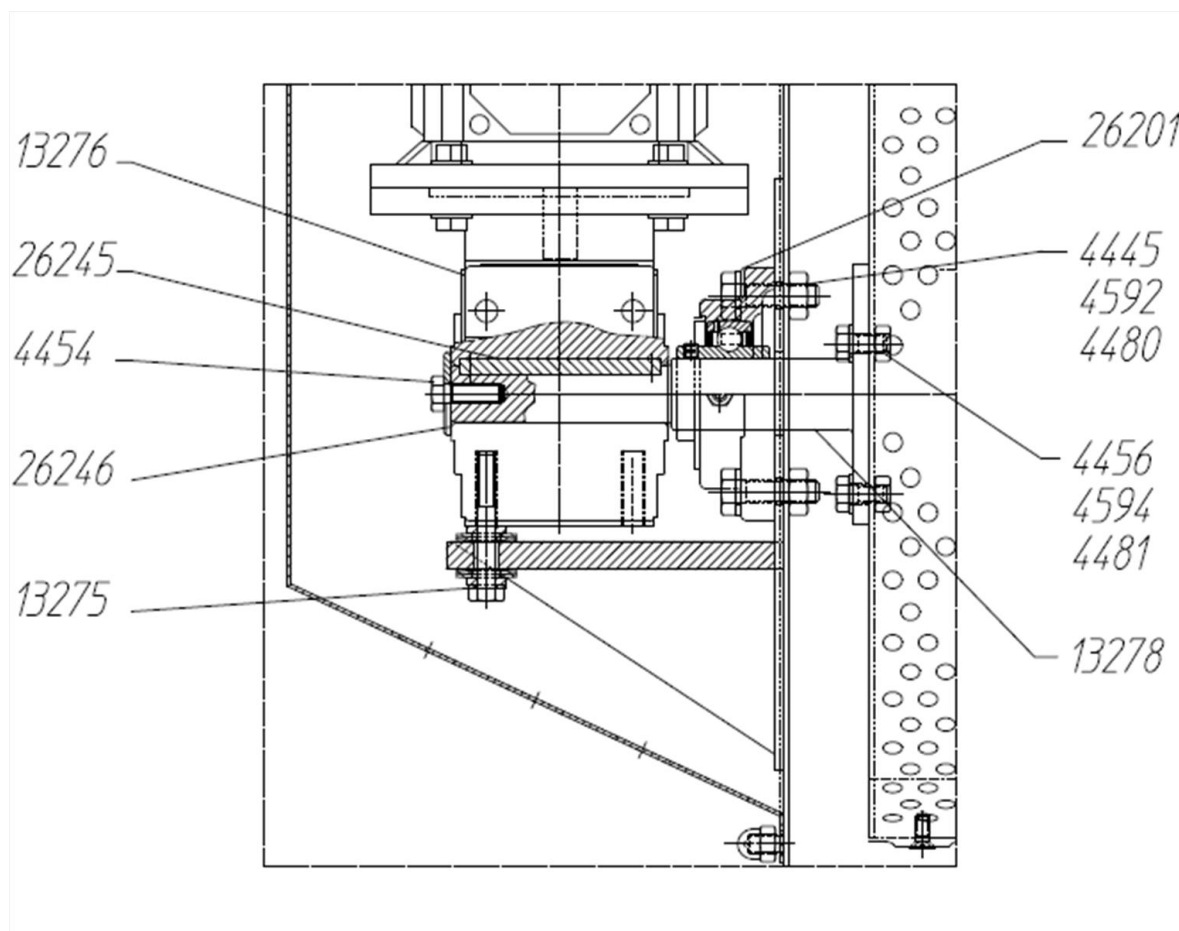
14.2 – Shutting system. Front Door



14.3 – Exploded view. Front door shut sensor



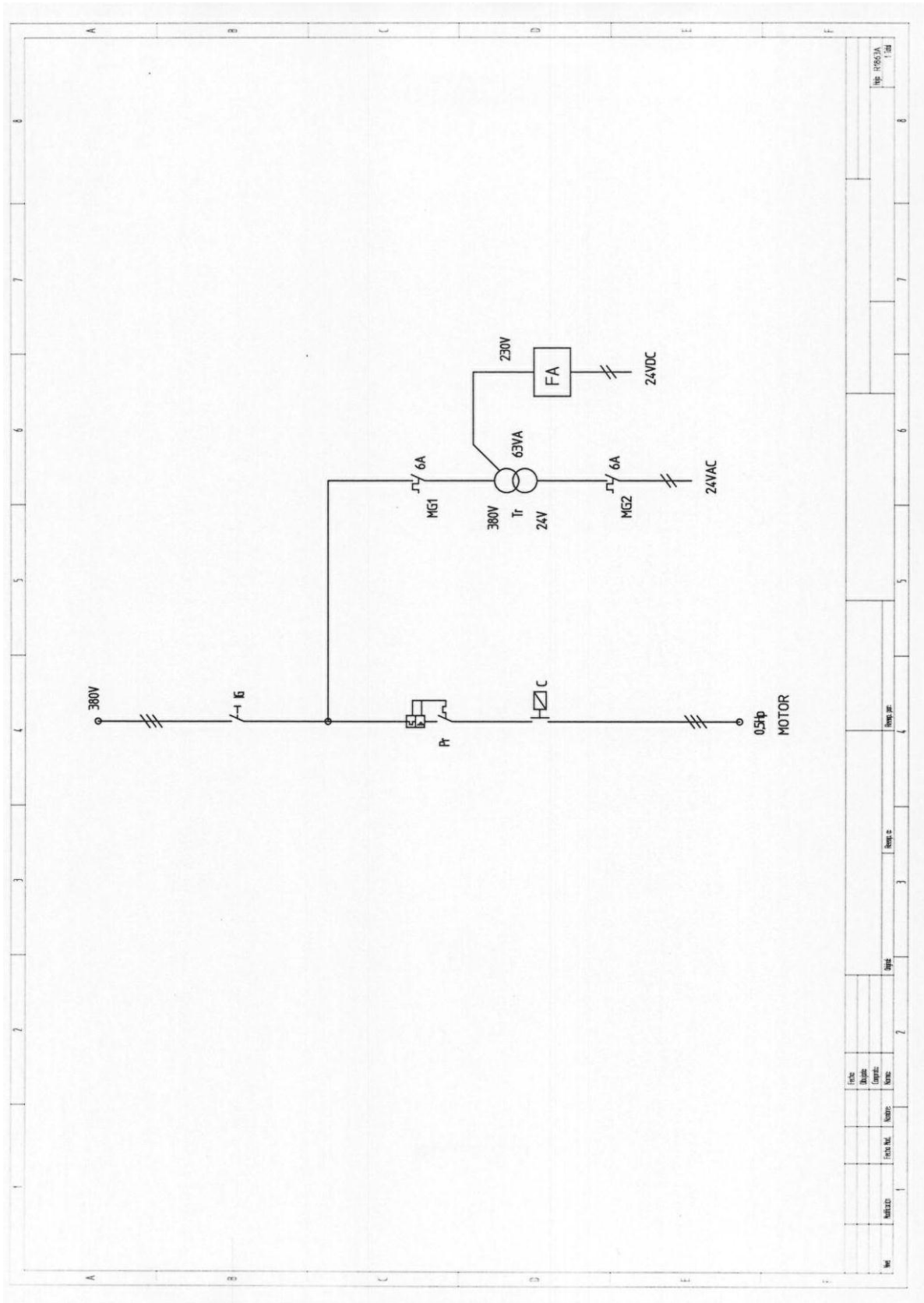
14.4 – Exploded view. Drive



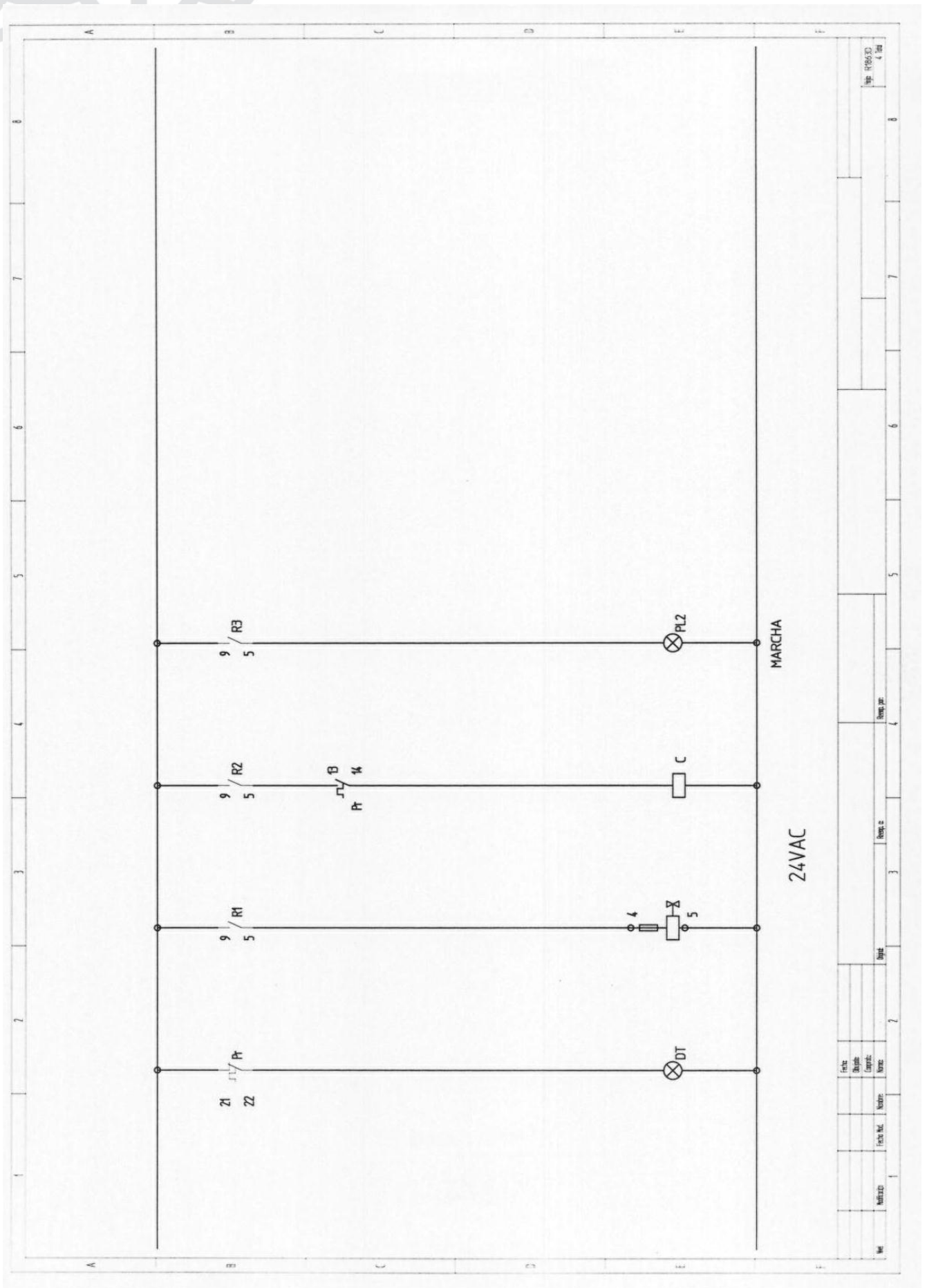
15 – LIST OF PARTS

Part Nr.	Units	Description	Material
1782	1	Hexagonal cap nut, M10 – DIN 1587	
1858	2	Hexagonal cap nut, M6 – DIN 1587	
1859	3	Hexagonal cap nut, M8 – DIN 1587	
2628	5	Flat washer, ø24/9 x 2 mm	
3885	1	Locking lever knob, ø35 mm – Ref.: PPP-1112	Plástico
4445	1	Hexagonal head screw M8 x 25 mm - DIN 933	
4454	1	Hexagonal head screw M8 x 25 mm - DIN 933	
4455	3	Tonillo cabeza hexagonal M8 x 20 mm - DIN 933	
4456	4	Tonillo cabeza hexagonal M8 x 16 mm - DIN 933	
4460	1	Hexagonal head screw M6 x 30 mm - DIN 933	
4477	4	Hexagonal nut, M16 - DIN 934	
4478	2	Hexagonal nut, M14 - DIN 934	
4480	4	Hexagonal nut, M10 - DIN 934	
4481	4	Hexagonal nut, M8 - DIN 934	
4503	1	Countersunk head screw, M6x10 - DIN 963	
4592	8	Flat washer A10,5 (M10) - DIN 125	
4593	2	Flat washer A6,4 (M6) - DIN 125	
4594	7	Flat washer A8,4 (M8) - DIN 125	
4599	2	Spring washer M14. - DIN 127 - (Grower)	
9357	9	Countersunk head screw, M6x12 - DIN 963	
9941	7	Threaded rod M8 x 10 mm - DIN 913 - (Allen)	AISI-304
10589	7	Spray nozzle G-3/8"	
10718	3	Front door screw	
10723	1	Rear cover	
10724	1	Drum front cover	
12734	4	Adjustable leg with rubber base M16.	
13275	1	Anchoring bolt fot gear motor	
13276	1	Gear motor 0.5cv MRIV40U03A-71B4 34.5RPM	
13278	1	Driving shaft	
13284	1	Lever shaft for Front door locking	
17867	1	Ball valve 1"	
18434	1	Lever arm for front door locking	
18635	3	Countersunk head screw, M8x110 - DIN 963	
19144	1	Coil for solenoid valve	
19145	1	Connector for solenoid valve coil	
19149	1	Solenoid valve housing G-1"	
20042	1	Eccentric bushing, ø30/20/15 x 40,5 mm	

Nº parte	Uds.	Descripción	Material
20043	3	Bushing for Connection of front door locking lever, ø30/15x10 mm	
20045	1	Bevelled washer, ø15/6,5 x 4 mm	
20051	2	Lower wheel, Drum rail, ø60/22,3 x 30 mm	Nylon
20052	1	Upper wheel, Drum rail, ø75/55/20,2 x 30 mm	Nylon
20057	2	Shaft for lower wheel, ø75/55/20,2 x 30 mm	
20066	1	Shaft ø15/12/M6x433 mm, for front door hingel	
20067	3	Drum inner paddle.	Nylon
20943	1	Water input pipe flange	
24003	1	Threaded rod for front door rail, M6x25 mm - Derecha - DIN 975	
25002	1	Rubber gasket, 20x20x1800 mm	
25246	1	Flat washer, ø35/9 x13 mm	
26201	1	Support with bearing	
26245	1	Key A-8x7x85 mm - DIN 6885	
29167	1	Inductive sensor IGT200 –IFM-	
OT-008581	1	Glove and apron washing machine drum	
OT-008598	1	Right support for front door	
OT-008599	1	Door locking handle	











LIST OF COMPONENTS - R-1863.F

Ref. in Schematics	Description	Brand	REFERENCIA
IG	Interruptor general Main switch	Telergon	T013 25 A+D714
Pr	Protector de motor Thermal/magnetic breaker	TEE	GV2ME06+GVAE11
C	Contactor	TEE	LC1D09B7
MG1 MG2	Thermal/magnetics	MG	24335
Tr	Transformador Transformer	Polylux	220/380/415/440-24V 63VA
FA	Fuente alimentación Power supply	TEE	ABL8MEM24012
PEG	Paro emergencia Emergency stop	TEE	ZB4BS54+ZB4BZ102+ ZBY9330
MS	Modulo seguridad Safety relay	TEE	XPSAC5121
RE1 . . . R3	Relés/relays	TEE	RXM4AB2B7+ RXZE2S114M
PL1	Piloto emergencia Emergency pilot lamp	TEE	ZB4BV053+ ZB4BVB5
PL2	Piloto marcha Run pilot lamp	TEE	ZB4BV033+ ZB4BVB3
PLC	Autómata Program Logic Controller	TEE	SR2B201BD
M	Pulsador marcha Run pushbutton	TEE	ZB4BP3+ZB4BZ101
Induc	Inductivo Inductive sensor	IFM	IGT200+ EVT005
DT	Piloto disparo térmico Thermal overload pilot lamp	TEE	ZB4BV043+ ZB4BVB4



ANNEX : Manual Zelic Logic 2. Configuration parameters

We detail as follows the configuration parameter section of the PLC. Please, request Adfood's technical department the complete user manual in digital format when needed.

PARAMETERS Menu

5

PARAMETERS Menu

Description This menu is used to enter and modify the application parameters directly on the screen using the smart relay keys. This function can be accessed in the two modes: **LD** and **FBD**, but the contents will be specific to the mode used.

If there are non-locked parameters to display they are listed in the window; otherwise a **NO PARAMETER** message appears.

LD mode Functions with parameters in LD mode:

- Auxiliary relays (see *Auxiliary Relays*, p. 93) (latching),
- Discrete Outputs (see *Discrete (DISCR) Outputs*, p. 98) (latching),
- Clocks (see *Clocks*, p. 134),
- Analog Comparators (see *Analog Comparators*, p. 129),
- Timers (see *Timers*, p. 101),
- Counters (see *Counters*, p. 111),
- Fast counter (see *Fast Counter*, p. 118).

Only those functions used in the program and with parameters are listed in the **PARAMETERS** menu.



FBD mode

Functions with parameters in FBD mode:

- Numerical Constant-Type Inputs,
- Clocks,
- Gain,
- Timers: TIMER A/C, TIMER B/H, TIMER Li,
- Counters: PRESET COUNT,
- Fast counter,
- CAM block.

To access the parameters of the FBD blocks, you must know and enter the block number. This number appears in the programming software on the wiring sheet at the top right corner of the block.

Only those functions used in the program and with parameters are listed in the **PARAMETERS** menu.

Parameter Modification

Parameter modification procedure:

Step	Action
1	Place the cursor over the PARAMETERS menu in the main menu (PARAMETERS flashing) and confirm by pressing the Menu/OK button. Result: The parameters window opens to the first parameter.
2	Select the function to modify. To access the required function, scroll through the function block numbers (navigation keys ▼ and ▲) until you reach the right one.
3	Select the parameter to modify. The ◀ and ▶ keys are used to place the cursor over the parameter to modify.
4	Modify the parameter using the + and - keys (▲ and ▼) of the contextual menu.
5	Confirm the modifications by pressing Menu/OK , which will open the confirmation window.
6	Confirm again twice by pressing Menu/OK to save. Result: The display returns to the INPUTS-OUTPUTS screen in RUN mode and to the MAIN menu in STOP mode.

Parameters in RUN Mode

It is therefore possible to modify parameters in RUN mode dynamically as long as they are not locked.

The modifications can be made:

- From the **PARAMETERS** (see *PARAMETERS Menu, p. 51*) menu,
- From the **MONITORING** (see *MONITORING Menu, p. 53*) (LD) menu: Move the pointer over the function to be modified using the navigation keys and open the parameters window from the contextual menu (**Shift** key).



This image shows a full page of blank graph paper. The grid consists of small, evenly spaced squares formed by thin gray lines. There are no margins, text, or other markings on the page.



adFOOD

**ADVANCED FOOD
SYSTEMS GROUP**

ROSER Construcciones Metàlicas S.A

Ctra Girona a Sant Feliu, n° 369-Km 11.5 -17244 -Cassà de la Selva (Girona) -España

Tel.: (34) 972 460 434 - Fax: (34) 972 463 700

Correo electrónico (e-mail) : roser@roser.es - Web <http://www.roser.es>